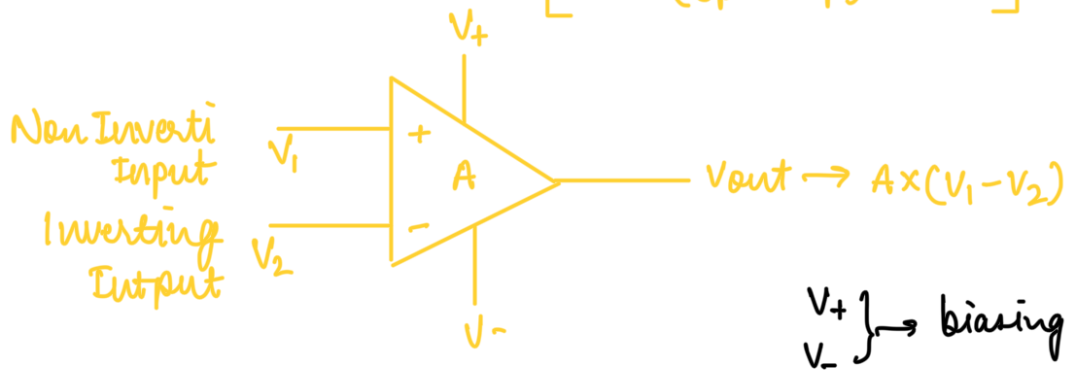
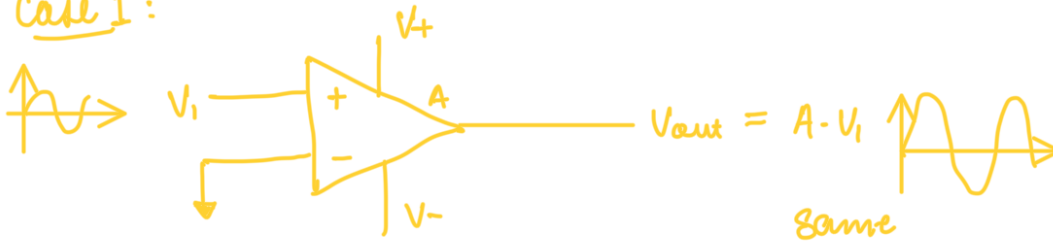


Operational amplifier (Op-Amp)

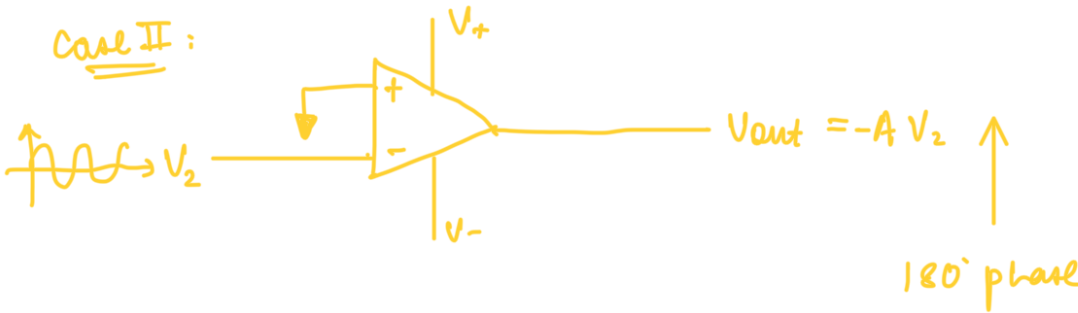


V_+
 V_- } $\rightarrow$ biasing voltage

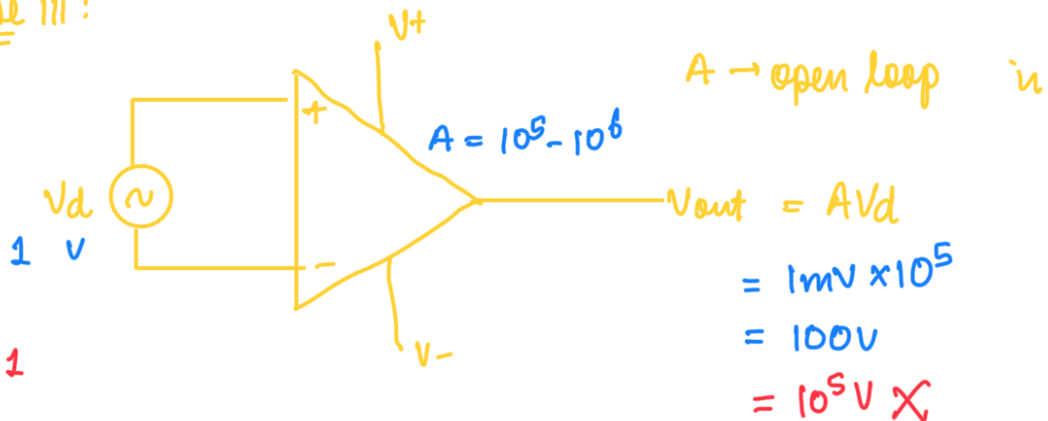
Case I:



Case II:

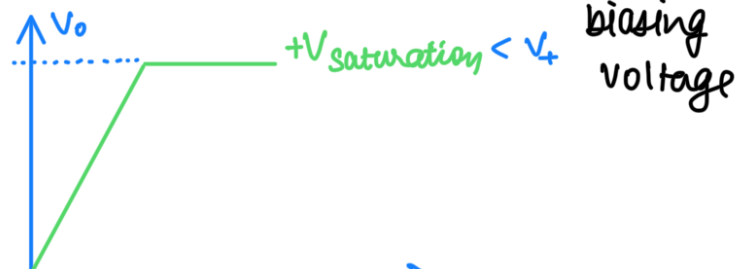


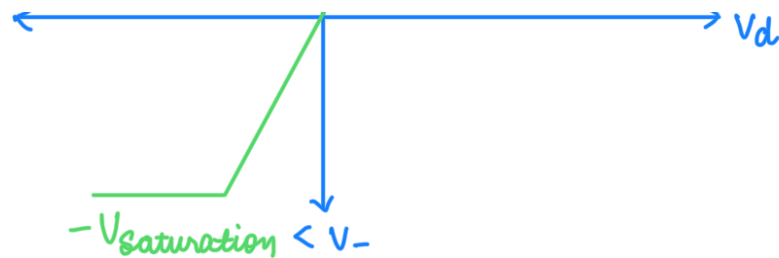
Case III:



the output voltage will not exceed

Voltage transfer curve of OP-AMP

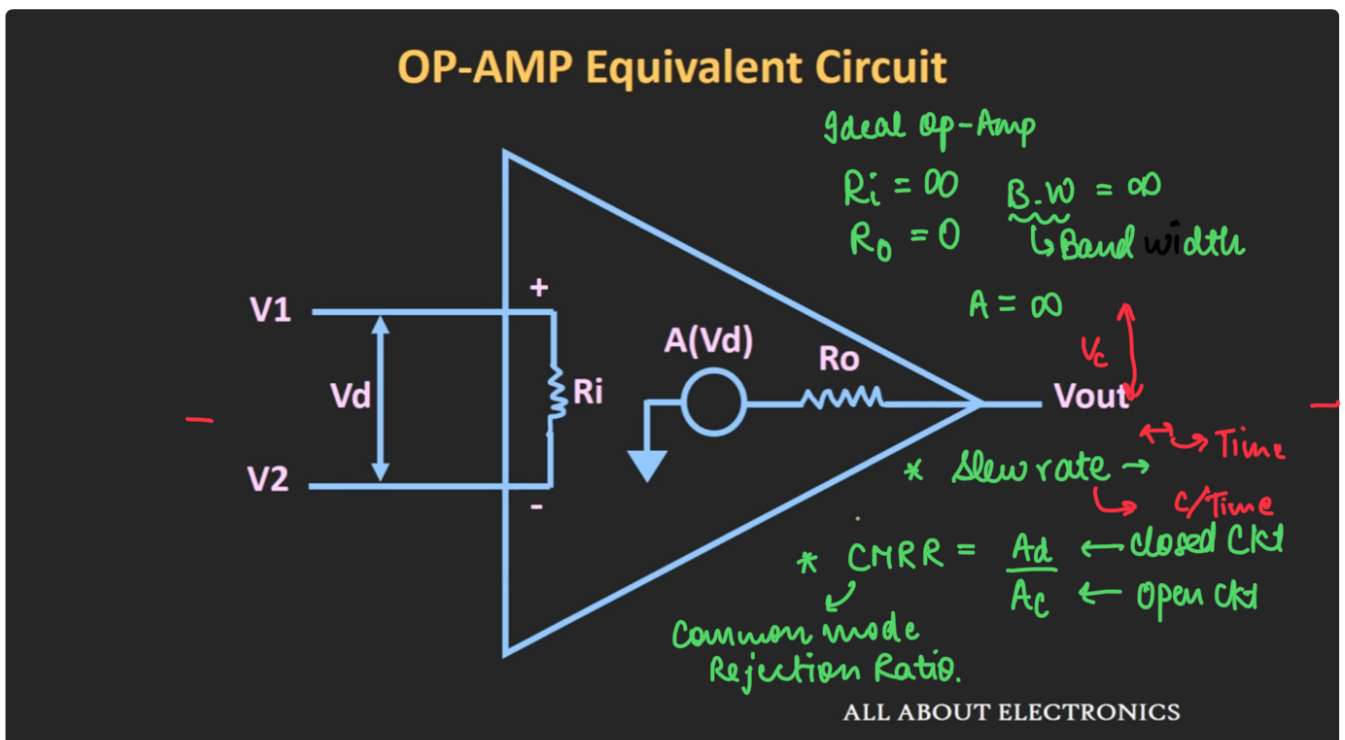




Uses:

- Active Filters
- Oscillator
- Waveform convertor
- A/D and DAC

OP-AMP Equivalent Ckt



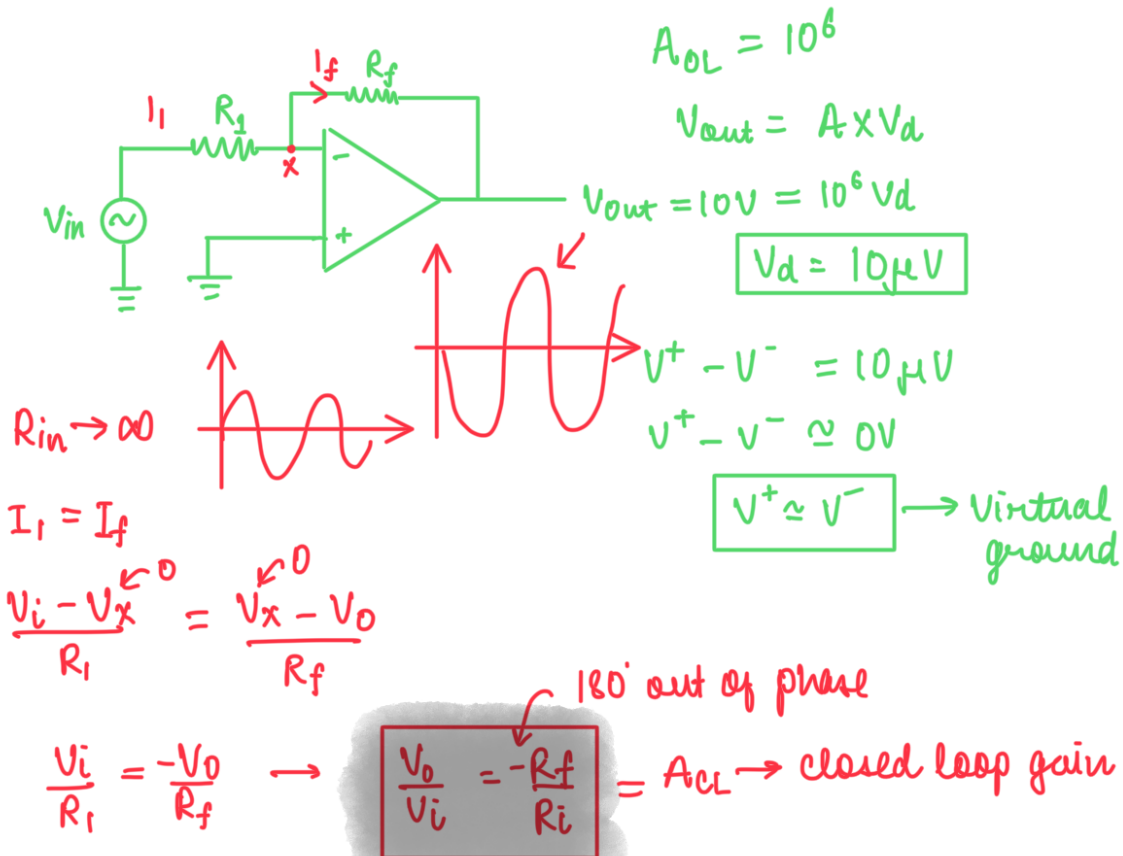
OP-AM 741 Specification:

Parameters	Value
Input Impedance R_i	$2M\Omega$
Output Impedance R_o	75Ω
open-loop Gain	105
Offset voltage	1mV
Slewrate	$0.5V/\mu s$

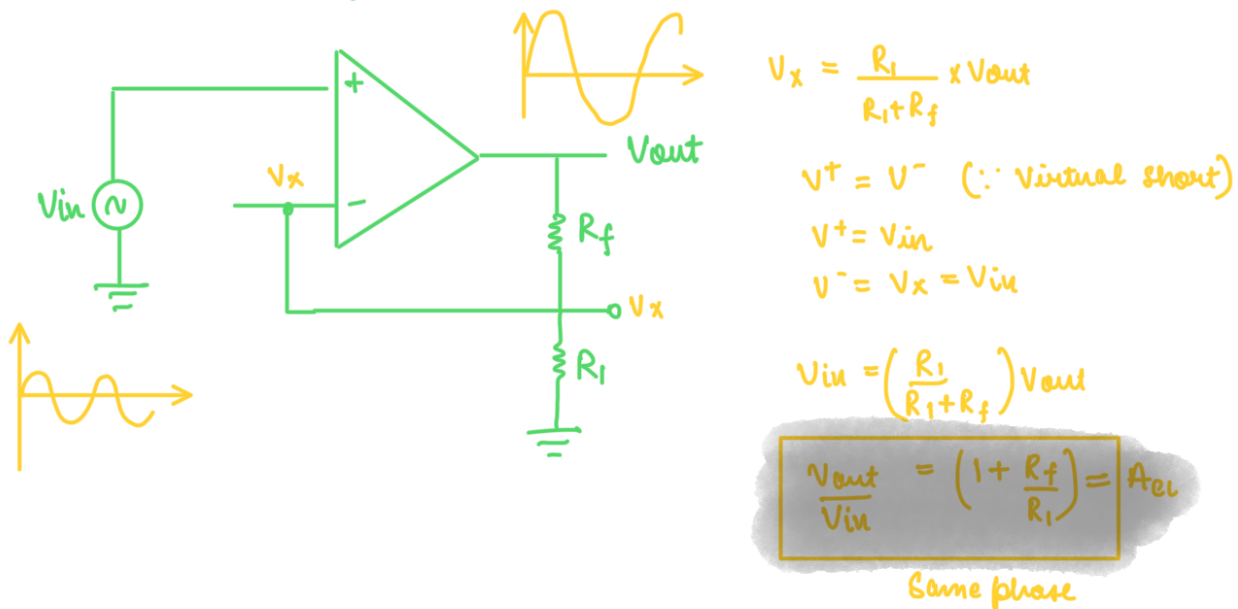
CMRR

70-90dB

Inverting OP-Amp:



Non Inverting OP-AMP



Inverting OPAMP

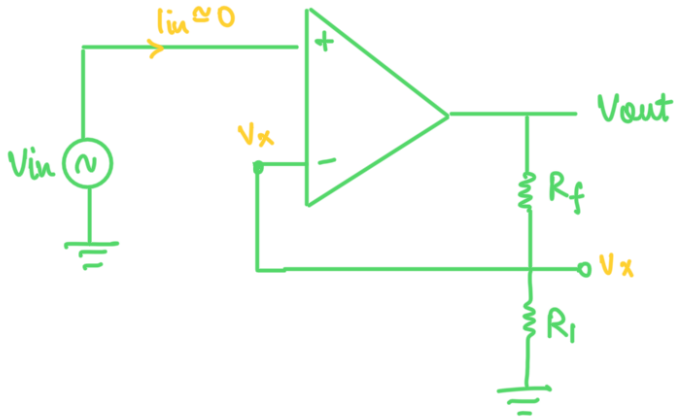


$$Z_{in} = \frac{V_{in}}{I_{in}}$$

$$I_{in} = \frac{V_{in}}{R_1} \rightarrow R_1 = \frac{V_{in}}{I_{in}} = Z_{in}$$



Non Inverting OP-AMP

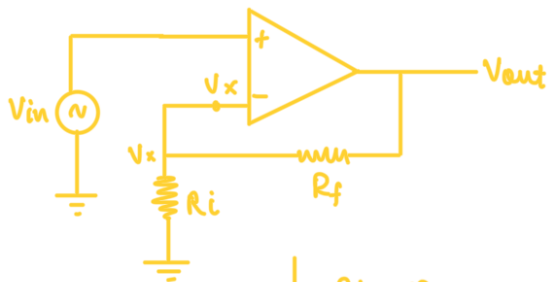


$$Z_{in} = \frac{V_{in}}{I_{in}}$$

$I_{in} = 0 \rightarrow$ Ideal OP-AMP

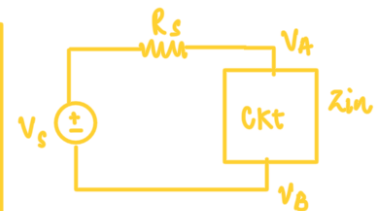
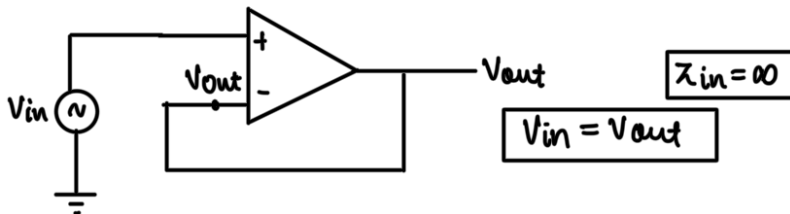
$$Z_{in} = \infty$$

OP-AMP as Buffer:



$$R_i = \infty$$

$$R_f = 0$$



$$Z_{in} = R_s$$

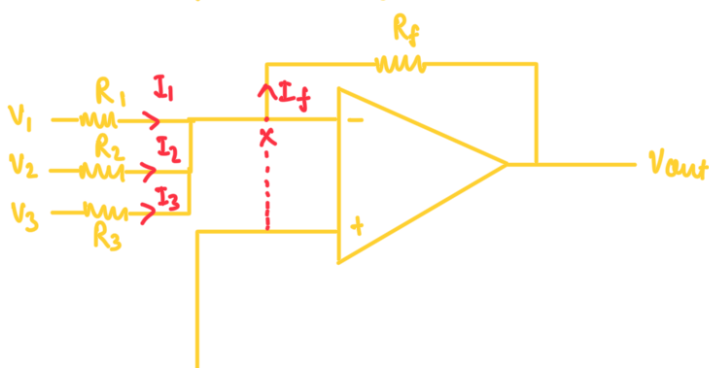
$$V = \frac{Z_{in}}{Z_{in} + R_s} \times V_s = \frac{V_s}{2}$$

$\therefore$ Z_{in} must be higher so voltage diff across ckt is zero

$$V_A - V_B = 0$$

Ideally

Inverting Summing Amplifier:



KCL at X

$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$V_{out} = -R_f \left[\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right]$$

$\text{if } R_1 \neq R_2 \neq R_3, \quad \frac{R_f}{R_1} \neq \frac{R_f}{R_2} \neq \frac{R_f}{R_3} \quad A \neq B \neq C$

$V_{out} = -[AV_1 + BV_2 + CV_3]$

$\rightarrow \text{Scaling}$

$\text{if } R_1 = R_2 = R_3 = R, \quad V_{out} = -\left(\frac{R_f}{R}\right)[V_1 + V_2 + V_3]$

$\rightarrow A_{cl}$

$\text{if } R_1 = R_2 = R_3 = R_f = R \quad V_{out} = -[V_1 + V_2 + V_3]$

$\rightarrow \text{Adder}$

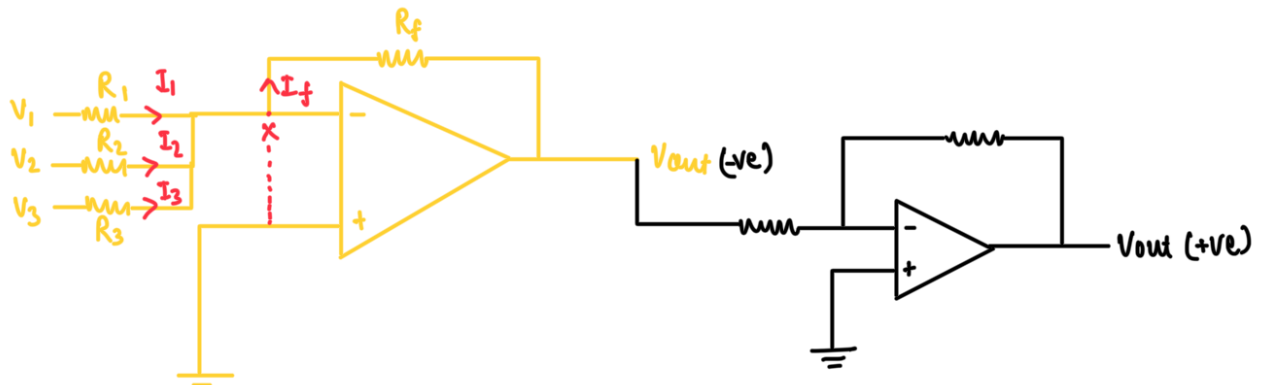
$\text{if } R_1 = R_2 = R_3 = R, \quad \frac{R_f}{R} = \frac{1}{n} \rightarrow \text{no. of Input (say 3)}$

$V_{out} = -\frac{1}{n}(V_1 + V_2 + V_3)$

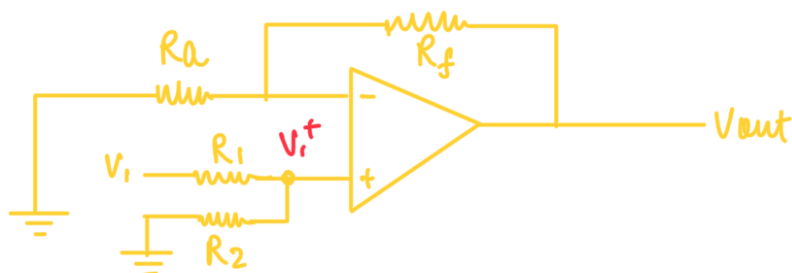
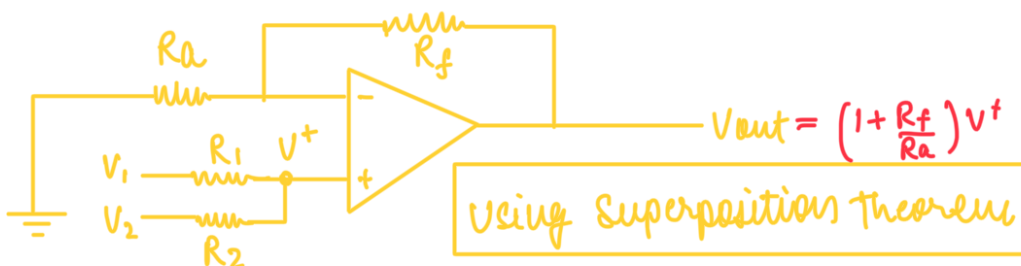
$= -\left(\frac{V_1 + V_2 + V_3}{3}\right)$

$\rightarrow \text{Averaging}$

To get +ve output we use inverting amplifier having $R_1 = R_f$:



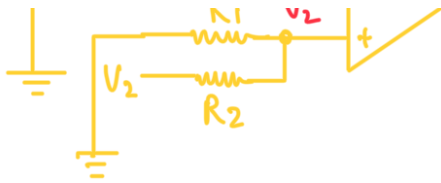
Non Inverting Summing Amplifier:



$$V_1^+ = \frac{R_2 \times V_1}{R_1 + R_2}$$



$$V_2^+ = \frac{R_1 \times V_2}{R_1 + R_2}$$



$$R_1 + R_2$$

$$V^+ = V_1^+ + V_2^+$$

$$V^+ = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) \left(\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2\right)$$

* if $R_1 = R_2 = R$

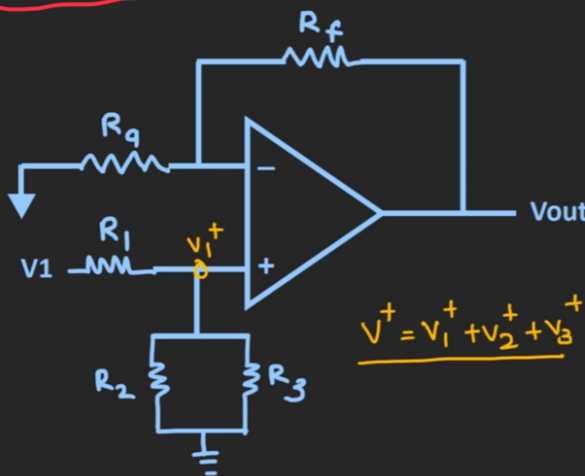
$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) \left(\frac{V_1 + V_2}{2}\right)$$

* if $R_1 = R_2 = R$ & $R_f = R_a$:

$$V_{out} = V_1 + V_2 \rightarrow \text{Adding}$$

Non-Inverting Summing Amplifier

3 Input



$$V_1^+ = \frac{(R_2 || R_3)}{R_1 + (R_2 || R_3)} \times V_1$$

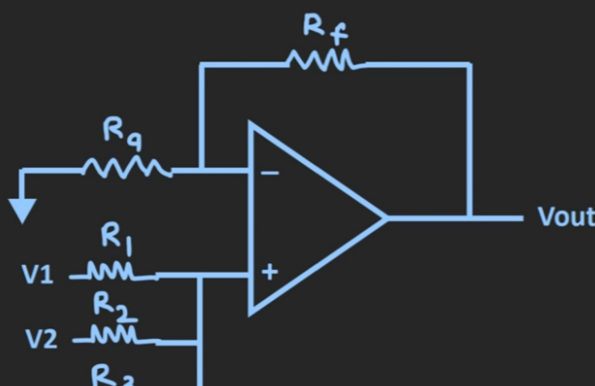
$$V_2^+ = \frac{(R_1 || R_3)}{R_2 + (R_1 || R_3)} \times V_2$$

$$V_3^+ = \frac{(R_2 || R_1)}{R_3 + (R_2 || R_1)} \times V_3$$

$$V^+ = V_1^+ + V_2^+ + V_3^+$$

ALL ABOUT ELECTRONICS

Non-Inverting Summing Amplifier



$$V^+ = \frac{V_1 + V_2 + V_3}{3}$$

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) V^+$$

$$= \left(1 + \frac{R_f}{R_a}\right) \left(\frac{V_1 + V_2 + V_3}{3}\right)$$

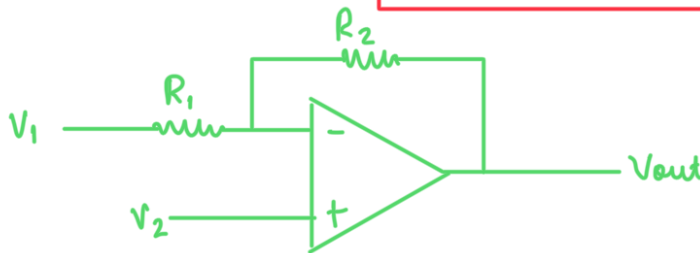
V3

$$1 + \frac{R_f}{R_a} = 3$$

$$= V_1 + V_2 + V_3$$

ALL ABOUT ELECTRONICS

Differential OP-Amp



$$V_{out} = V_{o1} + V_{o2}$$

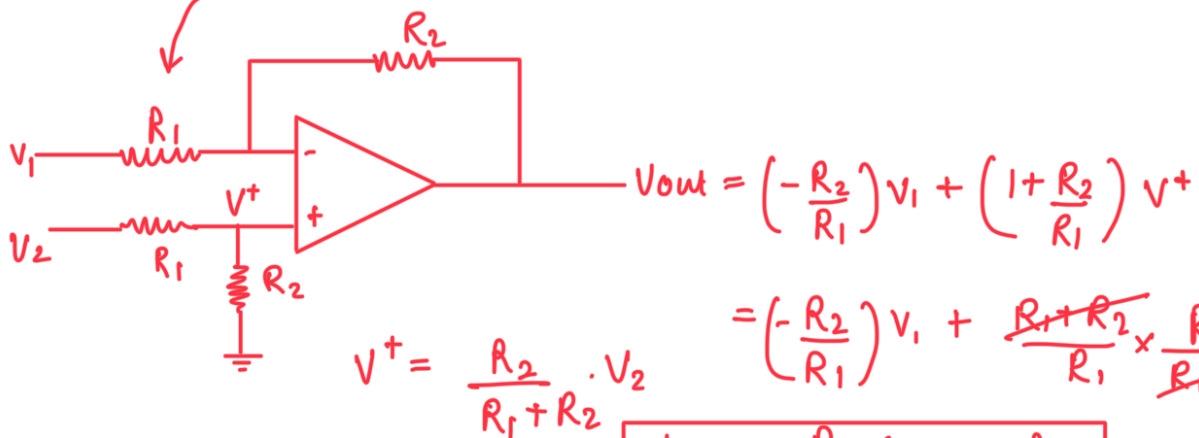
$$V_{out} = \left(-\frac{R_2}{R_1}\right)V_1 + \left(1 + \frac{R_2}{R_1}\right)V_2$$

$$V_{out} = K(V_2 - V_1)$$

$$\text{Let } K = \frac{R_2}{R_1}$$

$$1 + \frac{R_2}{R_1} = \frac{R_2}{R_1}$$

$$\frac{R_2 + R_1}{R_1} \times \frac{R_2}{R_1 + R_2} = \frac{R_2}{R_1}$$



$$V_{out} = \left(-\frac{R_2}{R_1}\right)V_1 + \left(1 + \frac{R_2}{R_1}\right)V^+$$

$$V^+ = \frac{R_2}{R_1 + R_2} \cdot V_2$$

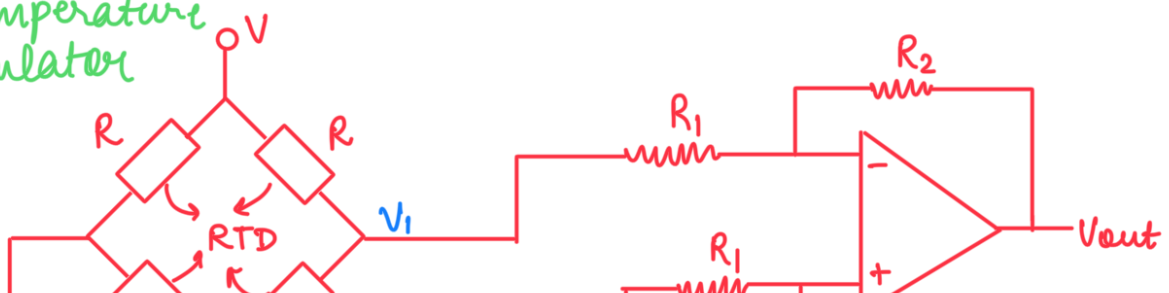
$$= \left(-\frac{R_2}{R_1}\right)V_1 + \frac{R_1 + R_2}{R_1} \times \frac{R_2}{R_1 + R_2} V_2$$

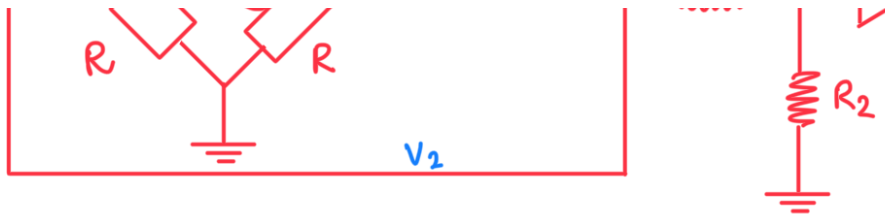
$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1)$$

Differential Amplifier

Application of Differential Amplifier:

* Temperature Regulator





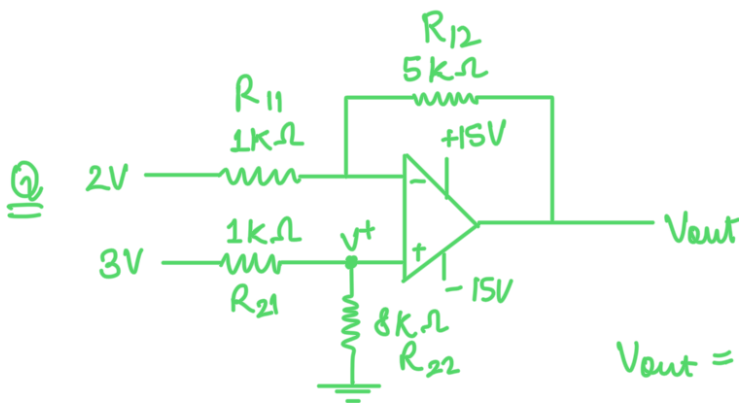
When the wheatstone bridge is balanced, the $V_1 = V_2$ and $V_{out} = 0$.

we can use V_{out} to control the temp of multiple rooms.

If the temperature changes, the Resistance of RTD changes which cause the V_1 & V_2 change and

$$V_1 \neq V_2$$

$V_{out} \neq 0 \rightarrow$ we can use to adjust temp.



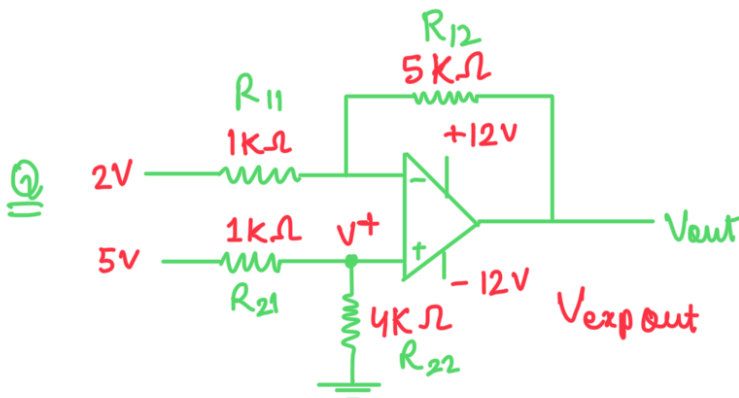
$$V_{out} = 2\left(-\frac{5}{1}\right) + \left(1 + \frac{5}{1}\right)V^+$$

$$= -10V + \frac{2 \times 8}{3}$$

$$= (-10 + 16)V$$

$$V^+ = \frac{8}{3} \times 3$$

$$V_{out} = 6V$$



$|V_{calculated}| > |biasing\ voltage|$
 $\therefore V_{out} = biasing\ voltage$

$$V_{expout} = V_{o1} + V_{o2}$$

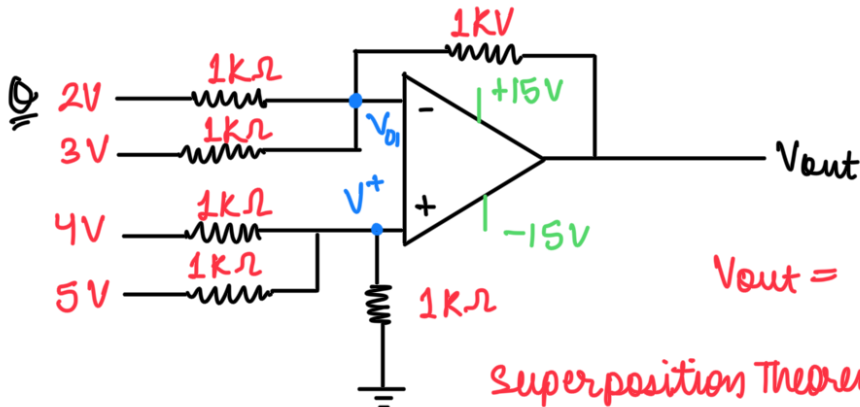
$$= -10V + 24V$$

$$= 14V < 12V \therefore V_{out} = 12V$$

$$* V_{o1} = 2\left(-\frac{5}{1}\right) = -10V$$

$$V_{O2} = \left(1 + \frac{5}{1}\right) V^+ \quad V^+ = \left(\frac{4}{4+1}\right) 5$$

$$* V_{O2} = 6 \times 4V = 24V$$



$$V_{out} = V_{O1} + V_{O2} = \boxed{4V} < 15V \quad \text{Ans} \quad \checkmark$$

Superposition Theorem:

$$* V_{O1} = 2 \left(\frac{-1}{1} \right) + 3 \left(\frac{-1}{1} \right) = -5V$$

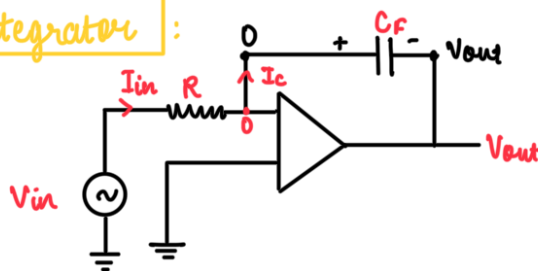
$$* V_{O2} = V^+ \left(1 + \frac{1}{1} \right) = 4.5(2) = 9V$$

Superposition Theorem:

$$V^+ = 4 \left(\frac{1}{1+1} \right) + 5 \left(\frac{1}{1+1} \right)$$

$$V^+ = 2 + 2.5 = \boxed{4.5V}$$

Integrator:



$$I_{in} = I_c \rightarrow \frac{V_{in} - 0}{R} = I_c$$

$$I_c = C \cdot \frac{dV_c}{dt} = \frac{V_{in}}{R}$$

$$\frac{V_{in}}{R} = C \cdot \frac{d(0 - V_{out})}{dt}$$

$$\frac{V_{in}}{R} = -C \cdot \frac{dV_{out}}{dt}$$

$$\boxed{-\frac{1}{RC} \int V_{in} dt = V_{out}}^*$$

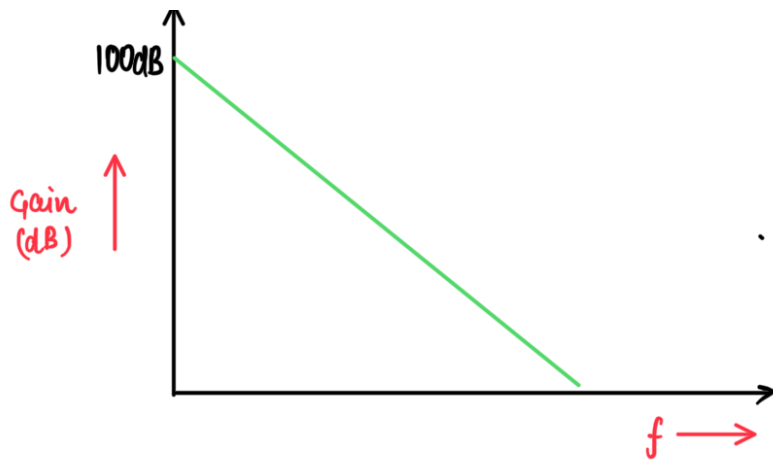
$$X_c = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega X_c}$$

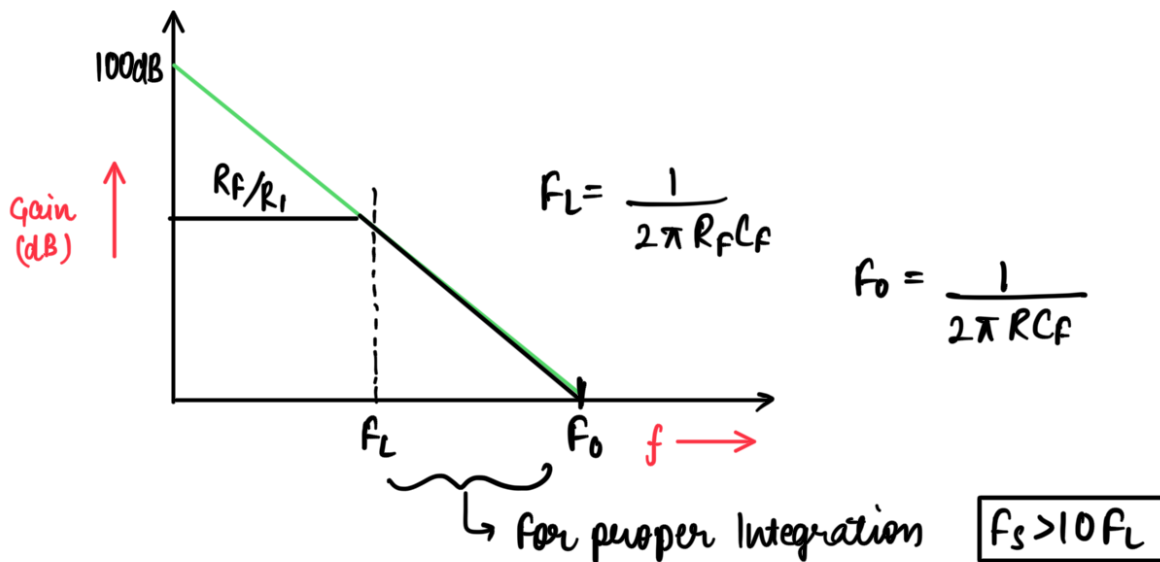
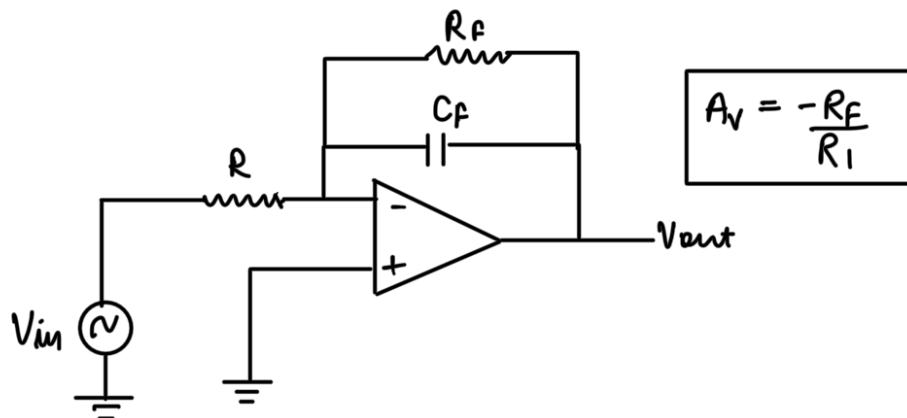
$$V_{out} = -\frac{X_c}{R} \cdot V_{in}$$

$$\boxed{A_v = -\frac{X_c}{R} = \frac{1}{2\pi RCf}}$$

Frequency Response of gain Integrator:



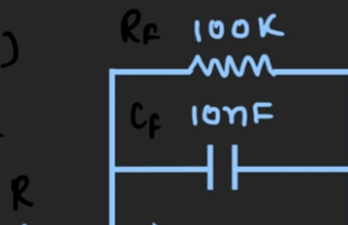
Practical Integrator using OP-Amp:



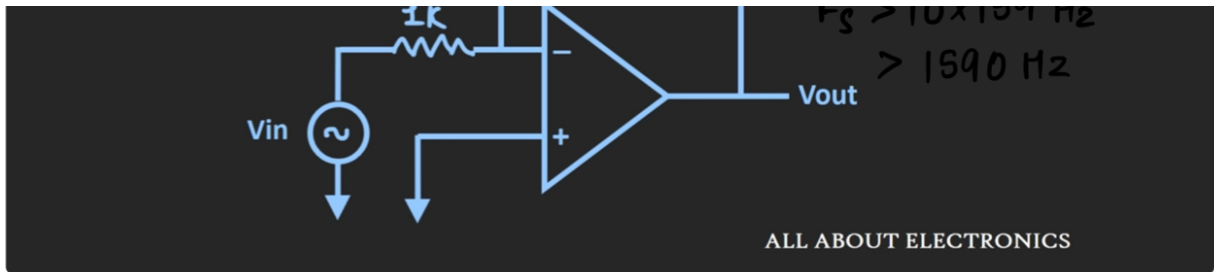
Example

Find Lower frequency limit of Integration for the circuit

① $V_{in} = \sin(2\pi 5000t)$
 $f_s = 5K > 1590Hz$



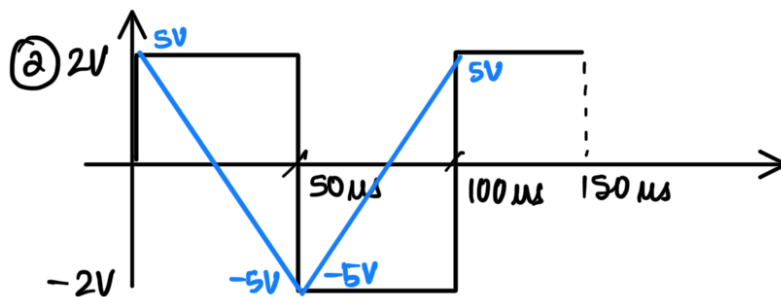
$F_L = \frac{1}{2\pi R_F C_F}$
 $= 159 Hz$



$$V_{out} = \frac{-1}{1k \times 10^{-8}} \cdot \int \sin(2\pi \cdot 5000t) dt$$

$$= \frac{10^{-5} \cos(2\pi \cdot 5000t)}{2\pi \cdot 5000}$$

$$V_{out} = 3.18 \cdot \cos(2\pi \cdot 5000t) \quad *$$



$$T = 100 \mu s$$

$$f = 10 \text{ KHz}$$

$$V_1(t) = 2V; \quad 0 < t < 50$$

$$V_2(t) = -2V; \quad 50 < t < 100$$

$$V_{o1} = \frac{-1}{R C_f} \int V_1(t) dt$$

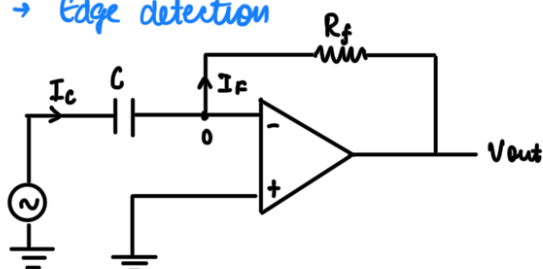
$$= -10^5 \cdot \int_0^{50 \mu s} 2 \cdot dt$$

$$= -10^5 \cdot 2 \cdot 50 \mu s$$

$$V_{o1} = -10V$$

OP-AMP - Differentiator :

Use $\rightarrow$ Find high frequency component of the signal
 $\rightarrow$ Edge detection

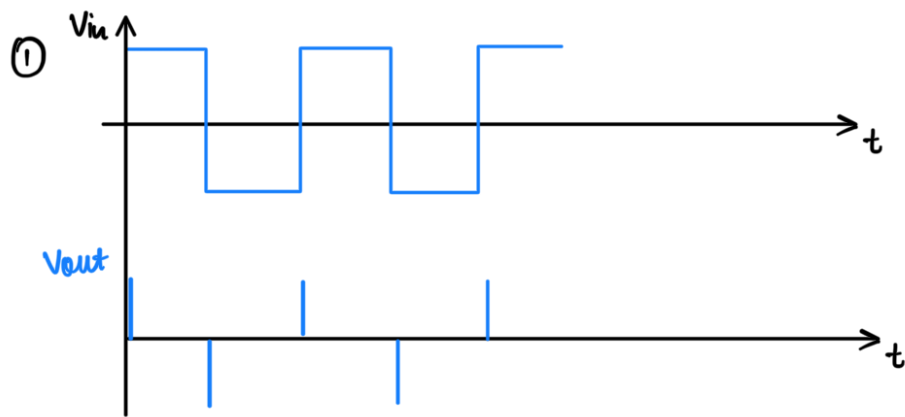


$$I_C = I_F$$

$$C \cdot \frac{dV_C}{dt} = \frac{0 - V_{out}}{R_F}$$

$$C \cdot \frac{dV_{in}}{dt} = -\frac{V_{out}}{R_F}$$

$$V_{out} = -R_F \cdot C \frac{dV_{in}}{dt}$$



② Sine wave $\rightarrow$ cosine wave

③ Triangular wave $\rightarrow$ square wave.